

Disproof of Conjecture 000000007668

The geometric mean of finite q-Pochhammer zeros

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October 5, 2026

The explicit source assertion. Both language versions define g_N as the nonnegative N th root of the product of the moduli of all N zeros of $(z; q)_N$, and assert, for each fixed real $0 < q < 1$, that

$$g_N = q^{-(N-1)/2}(1+q)^{-1}(1+O(q^{N/2})) \quad \text{as } N \longrightarrow \infty.$$

Here N is a positive integer, zeros are counted with algebraic multiplicity, and all exponents and divisions are real. The implicit constant and starting index in the O term may depend on q .

Meaning of the polynomial. We use the standard finite q-Pochhammer notation [1]:

$$P_{q,N}(z) = (z; q)_N = \prod_{j=0}^{N-1} (1 - zq^j), \quad z \in \mathbb{C}.$$

The source calls this a “partial q-binomial” without giving another formula. The displayed standard symbol supplies the convention used here. An alternative truncated series would require its own definition; it is not silently substituted for this polynomial.

Proposition 1. *For every real $0 < q < 1$ and every positive integer N , the geometric mean of all complex zero moduli of $P_{q,N}$ is exactly $q^{-(N-1)/2}$. Consequently the displayed relative-error assertion in the source is false.*

Proof. Each factor is a nonzero degree-one polynomial whose root is q^{-j} . The product has degree N , and its complete multiset of roots is therefore q^{-j} for $j = 0, \dots, N-1$, counting the multiplicities supplied by these factors. All these numbers are positive real numbers. Hence

$$\prod_{i=1}^N |z_i| = \prod_{j=0}^{N-1} q^{-j} = q^{-N(N-1)/2}, \quad g_N = \left(q^{-N(N-1)/2} \right)^{1/N} = q^{-(N-1)/2}.$$

Let $A_N = q^{-(N-1)/2}/(1+q)$. It is positive, and the source’s relative error is thus exactly

$$\frac{g_N}{A_N} - 1 = q.$$

If the asserted O bound held, there would be a real $C > 0$ and an integer $N_0 \geq 1$ such that $q \leq Cq^{N/2}$ for every $N \geq N_0$. But $q^{N/2} \rightarrow 0$ for fixed $0 < q < 1$, so this would imply $q \leq 0$, contradicting the hypothesis. $\square$

Relation to the whole conjecture. The displayed asymptotic formula is a necessary assertion in both language versions. Its negation disproves the conjunction under the declared standard notation. The later phrases do not specify the theta functions, Jensen mean, correction coefficient or referent of “this limit constant.” This argument does not assign definitions to those phrases or separately settle the stated transcendence refinement. Establishing the displayed formula alone would not prove those refinements; refuting it suffices for the stated conjunction.

Exact formal target. In `Conjecture7668.lean`, `qPochhammer` is a polynomial over $\mathbb{C}$. Its evaluation is proved to equal the displayed finite product. `rootModulusProduct` multiplies the norms in Mathlib’s actual `Polynomial.roots` multiset. The theorems `qPochhammer_roots` and `qPochhammer_roots_card` prove both the full multiset identity and its cardinality N . `geometricMean` uses `Real.rpow` with exponent $1/(N : \mathbb{R})$; nonnegativity and the N th-power identity are proved for $N > 0$. The totalized value at $N = 0$ is irrelevant to the eventual assertion. `RelativeAsymptotic q` is exactly

$$\exists C \in \mathbb{R}, C > 0 \wedge \exists N_0 \in \mathbb{N}, N_0 \geq 1 \wedge \forall N \geq N_0, \left| \frac{g_N}{A_N} - 1 \right| \leq Cq^{N/2}.$$

The lemma `relativeAsymptotic_iff_isBig0` proves equivalence to Mathlib’s `Asymptotics.IsBig0` along `Filter.atTop`. There is no uniformity requirement in q , finite cutoff, or restriction to a subsequence. The exponents use real subtraction and real division. `NecessaryClause` universally quantifies the preceding statement over all $0 < q < 1$. The theorem `conjecture7668_disproof` proves its negation; `not_relativeAsymptotic` and `not_relative_isBig0` give the stronger failure for every such q . The final logical lemma `no_completion` accepts the premise `FullSource` $\Rightarrow$ `NecessaryClause` and derives $\neg$ `FullSource`. This premise records the semantic implication explained above; it is not a formal definition of the source’s unspecified later refinements.

Verification and reproduction. Lean 4.19.0 and Mathlib v4.19.0 are pinned by the included toolchain and manifest. The unchanged proof was rebuilt in fresh independent projects, then both proof and inspection sources were replayed with warnings treated as errors. The audit covers all 29 authored declarations (8 definitions and 21 theorems), plus all 9 additional compiler-generated constants. Their types, definitions, axiom dependencies and logical dependency closures were inspected. Only `propext`, `Classical.choice` and `Quot.sound` occur; there are no admitted proofs, custom axioms, `native_decide`, or `unsafe/partial` proof dependencies. All nine dependency revisions and tracked sources were checked before and after replay. No numerical experiment or external mathematical computation is needed. The auxiliary programs inspect and reproduce the formal checks.

From the included `lean/` directory, after fetching the pinned packages and Mathlib cache, run `lake build`, then `lake env lean -DwarningAsError=true Conjecture7668.lean` and the same command for `Check.lean`. The submission’s `VERIFICATION.md` gives the full audit commands, exact file hashes, independent review and report checks. These are local validation records; maintainer acceptance is a separate decision.

References

- [1] NIST Digital Library of Mathematical Functions, equation 17.2.1, standard finite q-Pochhammer symbol.